

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 207 — Past Questions & Materials
by Renance

Course: GET 207

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 4

Levels: 200

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Paper 1 of 4 — GET 207 - Applied Mechanics

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

@8 DEPARTMENT OF MECHANICAL ENGINEERING

15th SEMESTER 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 207 - APPLIED MECHANICS

INSTRUCTION: ANSWER ALL QUESTIONS

TIME ALLOTTED: 2.5 HOURS

Question 1

What is a resultant force? (2 marks)

(a) Knowing that the resultant of the three forces shown in Figure Q1 is 300 N acting vertically downwards, determine the angle α . (12 marks)

Figure Q1

Question 2

(a) Identify and differentiate between the two types of force systems in terms of their dimensions (2 marks)

(b) The rectangular plate shown in Figure Q2 is supported by brackets at A and B and by a wire CD. Knowing that the tension in the wire is 300 N, determine the moment about A of the force exerted by the wire on point C.

(12 marks)

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Sen One

Question 3

7! (a) What is the principle of equilibrium?
n! (2 marks)

a A 120 kg crate is attached to the trolley beam system in equilibrium as shown

Figure Q3. If the distance $a = 1.5 \text{ m}$

determine

- (i) the tension in cable CD,
- (ii) the reaction at B (12 marks)

Figure Q3

Question 4

(a) Differentiate between the terms kinetics and kinematics (2 marks)

(b) The motion of a particle is defined by the relation $x = 6t^3 - 12t^2 + 3t + 2$ where x and t are expressed in meters and seconds, respectively. Determine the time, the position, and the velocity when the acceleration is equal to zero.

(2 marks)

Question 5

(a) Write short note on the following:

(i) Principle of work and energy (1 mark)

(ii) Principle of conservation of momentum — . (1 mark)

(b) The vehicle shown in Figure Q5 weighs

10000 kg. It is driven down the 5° incline road

at a speed of 66 km/h when the brakes are applied by

causing a constant total braking force (applied to

the road on the tires) of 2500 N. Determine (i)

the distance traveled by the automobile as

it comes to a stop using the work-energy equation

(ii) The deceleration of the vehicle when the

brakes are applied using Newton's law (iii) the

time it took the vehicle to come to rest using the

impulse-momentum equations. (12 marks)

Figure Q5

Paper 2 of 4 — GET 207 APPLIED MECHANICS 2024/2025

Level 200 • First Semester • PastQuestion

| FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
1st SEMESTER 2024/2025 ACADEMIC SESSION
COURSE CODE AND TITLE: GET 207 - APPLIED MECHANICS
INSTRUCTION: ANSWER ALL QUESTIONS
TIME ALLOTTED: 2.5 HOURS

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Question 1

What is a resultant force? (2 marks)

(a) Knowing that the resultant of the three forces shown in Figure Q1 is 300 N acting vertically downwards, determine the angle α . (12 marks)

Figure Q1 shows three forces acting on a point. The forces are 320 mm, 400 mm, and 300 mm. (a) Identify and differentiate between the two types of force systems in terms of their dimensions (2 marks)

(b) The rectangular plate shown in Figure Q2 is supported by brackets at A and B and by a wire CD. Knowing that the tension in the wire is 300 N, determine the moment about A of the force exerted by the wire on point C. (12 marks)

Figure Q2

Figure Q3

Figure Q4

Figure Q5

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Question 3

What is the principle of equilibrium (2 marks)

A 120 kg crate is attached to the trolley beam system in equilibrium as shown in Figure Q3. If the distance $a = 1.5$ m, determine the reaction at the support.

- (i) the tension in cable CD,
- (ii) the reaction at B (12 marks)

Question 4

- (a) Differentiate between the terms kinetics and kinematics (2 marks)
- (b) The motion of a particle is defined by the relation $x = 6t^3 - 2t^2 - 12t + 3t + 4$ where x and t are expressed in meters and seconds, respectively. Determine the acceleration when the acceleration is equal to zero.

(a) Write short note on the following:

- (i) Principle of work and energy (1 mark)
- (ii) Principle of conservation of momentum. (1 mark)

LEMMA (b) The vehicle shown in Figure Q5 weighs

10000 kg. It is driven down the 5° incline road at

65 km/h when the brakes are applied

causing a constant total braking force (applied by

the road on the tires) of 2500 N. Determine the

distance traveled by the automobile as

comes to a stop using the work-energy equation

(ii) The deceleration of the vehicle when the

brakes are applied using Newton's law (iii) the

time it took the vehicle to come to rest using the

impulse-momentum equations. (12 marks)

Figure Q5

2) Pay

Paper 3 of 4 — GET 207 APPLIED MECHANICS 2024 2025

Level 200 • First Semester • PastQuestion

| FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

| 1 SEMESTER 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 207 - APPLIED MECHANICS

INSTRUCTION: ANSWER ALL QUESTIONS

TIME ALLOTTED: 2.5 HOURS

ee —

Question 1

: . iw a» > e Ss i

What is a resultant force? (2 marks)

(a) Knowing that the resultant of the three forces shown in Figure Q1 is 300 N acting vertically downwards, determine the angle α . (12 marks)

4 150N Figure Q1

390 mn (a) Identify and differentiate between the two types of force systems in terms of their dimensions (2 marks)

(b) The rectangular plate shown in Figure Q2 is supported by brackets at A and B and by a wire CD. Knowing that the tension in the wire is 300 N, determine the moment about A of the force exerted by the wire on point C.

| wuestion 2

(12 marks)

Figure Q2

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Question 3

What is the principle of equilibrium (2 marks)

A 120 kg crate is attached to the trolley beam system in equilibrium as shown

Figure Q3. If the distance $a = 15\text{N}$

is to be determined

(i) the tension in cable CD,

(ii) the reaction at B (12 marks)

Question 4

(2 marks)

(a) Differentiate between the terms kinetics and kinematics

— $2t^3 - 12t^2 + 3t + 4$

by the relation $x = 6t^3$

Determine t

in seconds, respectively.

When the acceleration is equal to zero.

(b) The motion of a particle is defined

where x and t are expressed in m

time, the position, and the velocity w

is

Question 5 |

(a) Write short note on the following:

(1 mark)

(i) Principle of work and energy

(ii) Principle of conservation of momentum (1 mark)

(b) The vehicle shown in Figure Q5 weighs

10000 kg. It is driven down the 5° incline road at

a speed of 66 km/h when the brakes are applied by

causing a constant total braking force (applied to

the road on the tires) of 2500 N. Determine (i)

the distance traveled by the automobile as

it comes to a stop using the work-energy equation

(ii) The deceleration of the vehicle when the

brakes are applied using Newton's law (iii) the

time it took the vehicle to come to rest using the

impulse-momentum equations, (12 marks)

Figure Q5 ,

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Paper 4 of 4 — GET 207 Applied Mechanics 2025

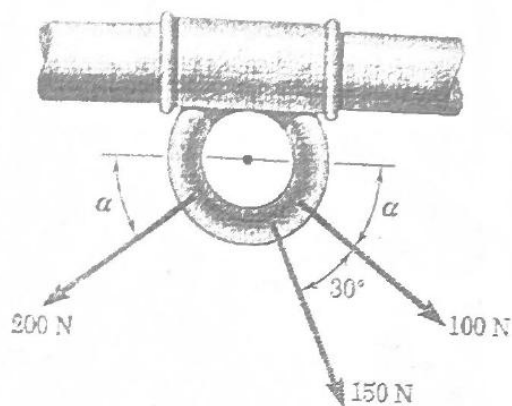
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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
1ST SEMESTER 2024/2025 ACADEMIC SESSION
COURSE CODE AND TITLE: GET 207 - APPLIED MECHANICS
INSTRUCTION: ANSWER ALL QUESTIONS
TIME ALLOTTED: 2.5 HOURS

Question 1

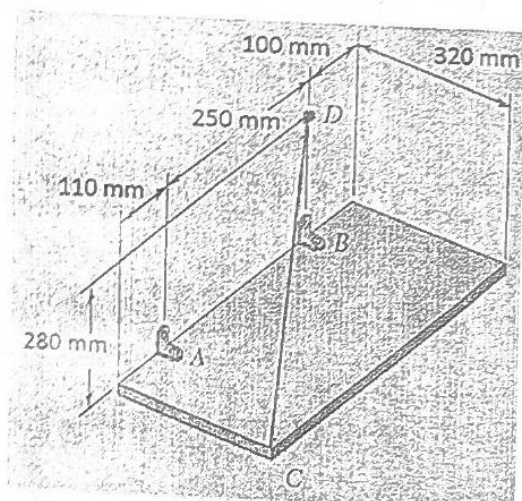


What is a resultant force? (2 marks)

(a) Knowing that the resultant of the three forces shown in Figure Q1 is 300 acting vertically downwards, determine the angle α . (12 marks)

Figure Q1

Question 2



(a) Identify and differentiate between the two types of force systems in terms of their dimensions (2 marks)

(b) The rectangular plate shown in Figure Q2 is supported by brackets at A and B and by a wire CD. Knowing that the tension in the wire is 300 N, determine the moment about A of the force exerted by the wire on point C. (12 marks)

Figure Q2